Chapter 14. Applications to Group Theory

Covering spaces of graphs, fundamental groups of graphs, subgroups of free groups

Contents

14.1 Covering spaces of a graph **2**

14.2 The fundamental group of a graph **2**

14.3 Subgroups of free groups **4**

14.4 Selected exercises **4**

Free groups are fundamental groups of graphs, so covering theory proves algebraic facts about them. (Not in the source PDF; results are named, not numbered.)

14.1 Covering spaces of a graph

Motivation. Covering spaces of graphs are again graphs.

A **linear graph** is a space X that is the union of arcs A_α (**edges**), any two meeting in at most a common end point (a **vertex**), whose topology is coherent with the edges: $C \subset X$ is closed iff $C \cap A_\alpha$ is closed in A_α for every α. A **subgraph** is a union of edges and vertices; it is closed. Fix for each edge a homeomorphism with $[0,1]$. For a subgraph S, the **open star** $\mathrm{St}(S)$ consists of S and the points of edges not in S at parameter distance less than $\frac{1}{2}$ from an end point lying in S; it is open by coherence. Since I is compact Hausdorff, $X \times I$ is coherent with the $A_\alpha \times I$ (quotient map $\times$ identity of a locally compact Hausdorff space is a quotient map).

Lemma 14.1 (graphs are nice)

A linear graph X is Hausdorff and locally path connected; every compact subset meets the interiors of only finitely many edges; and each subgraph S is a deformation retract of $\mathrm{St}(S)$. In particular X is semilocally simply connected.

Proof. Edge interiors and the stars $\mathrm{St}_\epsilon(v)$ (points within parameter distance ϵ of v) are open by coherence and path connected; they separate points and give local path connectedness. For compact C, pick one point of C in each edge interior where possible; every subset of the resulting D meets each edge in at most one point, so D is closed discrete in C, hence finite. Pushing each half-open segment of $\mathrm{St}(S) - S$ linearly to its end in S is continuous on each $(A_\alpha \cap \mathrm{St}(S)) \times I$, hence (coherence) a deformation retraction of $\mathrm{St}(S)$ onto S. Finally $\mathrm{St}(v)$ and small open arcs are contractible, so loops in them are nulhomotopic. ■

Theorem 14.2 (coverings of graphs)

If $p : E \to X$ is a covering map and X a linear graph, then E is a linear graph whose edges are the liftings of the edges of X, and p maps each edge of E homeomorphically onto an edge of X.

Proof. Parametrize an edge A by an injective path $f : I \to A$. For $e \in p^{-1}(f(0))$ the lifting $\tilde{f}$ from e is injective (as $p \circ \tilde{f} = f$ is), so its image is an arc mapped homeomorphically onto A. By uniqueness of liftings each point of $p^{-1}(A)$ lies on exactly one lifted arc; lifts of different edges meet at most in common end points over vertices, and two lifts of the same edge are disjoint. Coherence: let $D \subset E$ meet each lifted edge in a closed set, $e \in E$. Take an evenly covered $U \ni p(e)$ that is a small vertex star or open arc, with slice $V \ni e$. Each segment of U along one edge is connected, so lifts into V inside one lifted edge; hence $p(D \cap V)$ meets each segment in a closed set, so is closed in U (coherent with the segments), and $D \cap V$ is closed in V. Thus $E - D$ is open. ■

14.2 The fundamental group of a graph

Motivation. Collapsing a maximal tree shows π_1 of a graph is free, one generator per edge outside the tree.

A **tree** is a connected linear graph containing no simple closed curve.

Lemma 14.3 (finite trees)

A finite tree with $n \geq 1$ edges has a vertex lying on only one edge; it deformation retracts onto a vertex; and if it has V vertices then it has $V - 1$ edges.

Proof. Take a maximal arc $e_1 \cup \cdots \cup e_k$ of distinct edges (the tree is finite). Its last end point w lies on no other edge, which would extend the arc or close a simple closed curve. Removing e_k (but its other end) leaves a tree T' with $n - 1$ edges and one fewer vertex, and collapsing e_k linearly deformation retracts T onto T'. Induct. ■

Theorem 14.4 (trees are simply connected)

Every tree T is simply connected.

Proof. A loop f in T meets finitely many edge interiors (Lemma 14.1); the component $Y_0 \supset f(I)$ of the union of these edges is a finite tree, so f is nulhomotopic in $Y_0 \subset T$ (Lemma 14.3). ■

Lemma 14.5 (maximal trees)

Every connected linear graph X contains a maximal tree, and a maximal tree contains every vertex of X.

Proof. A union of a chain of trees is connected and has no simple closed curve (a compact curve lies in finitely many edges, hence in one member). Zorn gives a maximal tree T. If a vertex is not in T: were every edge with an end in T to have both ends in T, the subgraph W of T's vertices and these edges would be clopen, so $W = X$, a contradiction. So some edge A has exactly one end in T, and $T \cup A$ is a tree (a simple closed curve through the new end needs two edges there), contradicting maximality. ■

Theorem 14.6 (fundamental group of a graph)

Let X be a connected linear graph, T a maximal tree, x_0 a vertex, and $\{A_\alpha\}$ the edges not in T. For each α let f_α be a loop at x_0 that runs in T to one end of A_α, across A_α, and back in T. Then $\pi_1(X, x_0)$ is free with free generators $\{[f_\alpha]\}$.

Proof.

Step 1. $X = T \cup A_1 \cup \cdots \cup A_n$, induction on n ($n = 0$: Theorem 14.4). Let $Y = T \cup A_1 \cup \cdots \cup A_{n-1}$, $a \in \operatorname{int} A_n$, p, q the ends of A_n, γ the arc in T from p to q, $C = \gamma \cup A_n$ (a simple closed curve), $U = X - a$, $V = \operatorname{St}(C)$. U deformation retracts onto Y (push the halves of $A_n - a$ to p, q; pasting lemma on $Y \times I$, $(A_n - a) \times I$); V onto C and $U \cap V$ onto the arc $C - a$ (Lemma 14.1). Corollary 11.25 gives $\pi_1(X, p) \cong \pi_1(Y, p) * \pi_1(C, p)$; the first factor is free on the loops through $A_1, \ldots, A_{n-1}$ (induction), the second is $\mathbb{Z}$ on the loop around C, so Theorem 11.19 gives the claim at p; moving the base point along T turns these loops into the f_α.

Step 2. general case. A loop or path homotopy has compact image, so lies in $X_F = T \cup \bigcup_{\alpha \in F} A_\alpha$ for a finite F. Hence the $[f_\alpha]$ generate, and a nonempty reduced word in them equal to 1 (or a finite-order $[f_\alpha]$) would already occur in some X_F, contradicting Step 1. ■

Corollary 14.7 (Euler characteristic)

If X is a finite connected linear graph with V vertices and E edges, $\pi_1(X)$ is free of rank $E - V + 1 = 1 - \chi(X)$, where $\chi(X) = V - E$.

Proof. A maximal tree contains all V vertices (Lemma 14.5), hence has $V-1$ edges (Lemma 14.3), leaving $E-V+1$ edges outside it. ■

Consequently χ of a finite connected graph depends only on π_1 (by Corollary 11.22), and is unchanged by subdivision.

14.3 Subgroups of free groups

Motivation. A subgroup of π_1 of a graph is π_1 of a covering graph, so subgroups of free groups are free.

Subdividing each circle into three arcs makes a wedge of circles a linear graph; by Lemma 11.30 and Theorem 11.29 every free group is π_1 of one.

Theorem 14.8 (Nielsen–Schreier)

Every subgroup H of a free group F is free.

Proof. Let X be a wedge of circles with $\pi_1(X, x_0) \cong F$. X is path connected, locally path connected and semilocally simply connected (Lemma 14.1), so Theorem **??** gives a covering $p : E \to X$ with $p_*\pi_1(E, e_0) = H$. As p_* is injective, $H \cong \pi_1(E, e_0)$. E is a connected linear graph (Theorem 14.2), so $\pi_1(E, e_0)$ is free (Theorem 14.6). ■

Theorem 14.9 (Schreier index formula)

If F is free of rank n and H has finite index k in F, then H is free of rank $k(n-1)+1$.

Proof. In the proof above, the fibre $p^{-1}(x_0)$ is in bijection with the right cosets of H (Theorem **??**), so p is k-sheeted: each vertex and each edge of X has exactly k preimages. The subdivided wedge X has $2n+1$ vertices and $3n$ edges, so $\chi(X) = 1-n$ and $\chi(E) = k(1-n)$. By Corollary 14.7, $\pi_1(E)$ has rank $1-\chi(E) = k(n-1)+1$. ■

Example 14.1

(a) F free on a, b is π_1 of the figure eight X. The integer grid in $\mathbb{R}^2$ covers X (Example **??**); the integer translations act transitively on fibres, so the covering is regular with $F/H \cong \mathbb{Z}^2$, $H = p_*\pi_1(E)$. As F/H is abelian, $H \supset [F, F]$, and the surjection $\mathbb{Z}^2 \cong F/[F, F] \to F/H \cong \mathbb{Z}^2$ is injective (determinant ± 1), so $H = [F, F]$. The comb (x-axis and all vertical lines) is a maximal tree with infinitely many edges outside, so $[F, F]$ is free of infinite rank.
(b) $\ker(F \to \mathbb{Z}/2)$, $a, b \mapsto 1$, has index 2, hence rank $2(2-1)+1 = 3$: a subgroup of a free group can have larger rank.

14.4 Selected exercises

Motivation. Standard consequences of the theorems above (the source has no exercises here), with our own solutions.

Exercise 14.1 (recognizing finite trees)

Let X be a finite connected linear graph with V vertices and E edges.
(a) Show that $E \geq V - 1$, with equality iff X is a tree.
(b) Show that X is a tree iff $\pi_1(X)$ is trivial.

Solution [Supplement].

(a) A maximal tree T contains all V vertices (Lemma 14.5), so has $V-1$ edges (Lemma 14.3; $V = 1$ if none). Hence $E \geq V - 1$; X a tree $\Rightarrow T = X$, $E = V - 1$; $E = V - 1 \Rightarrow$ every edge lies in T, so $X = T$.

(b) Trees are simply connected (Theorem 14.4). If $\pi_1(X) = 1$, its rank $E-V+1$ is 0 (Corollary 14.7), so X is a tree by (a). ■

Exercise 14.2 (free groups are torsion free)

Show that every element $g \neq e$ of a free group F has infinite order.

Solution [Supplement].

Step 1. (Free on $S \neq \emptyset \Rightarrow$ infinite.) For $s \in S$, Lemma 11.18 gives $\phi : H \to \mathbb{Z}$ with $\phi(s) = 1$, $\phi(s') = 0$ $(s' \neq s)$; $\phi(s^k) = k$, so the s^k are distinct.

Step 2. If $g \neq e$ had finite order, $\langle g \rangle$ would be finite, nontrivial, and free on some $S \neq \emptyset$ (Theorem 14.8), contradicting Step 1. ■

Exercise 14.3 (free subgroups of every finite rank)

Let F be free on $\{a, b\}$. Show that for every $n \geq 1$, F has a subgroup that is free of rank n.

Solution [Supplement].

Step 1. $(n = 1.)$ a has infinite order (Exercise 14.2, Step 1), so $\langle a \rangle \cong \mathbb{Z}$ is free on $\{a\}$.

Step 2. $(n \geq 2.)$ Let $k = n - 1$. Lemma 11.18 gives an onto $\phi : F \to \mathbb{Z}/k$, $a \mapsto 1$, $b \mapsto 0$, so $H = \ker \phi$ has index k.

Step 3. By Theorem 14.9, H is free of rank $k(2-1)+1 = n$. ■

Exercise 14.4 (subgroups of index 2)

Let F be free of rank $n \geq 1$. Show that F has exactly $2^n - 1$ subgroups of index 2, and that each is free of rank $2n - 1$.

Solution [Supplement].

Step 1. Index 2 $\Rightarrow$ normal ($gH = Hg = F - H$ for $g \notin H$), so $H = \ker(F \to F/H \cong \mathbb{Z}/2)$; conversely kernels of surjections $F \to \mathbb{Z}/2$ have index 2.

Step 2. A surjection $\phi : F \to \mathbb{Z}/2$ is determined by its kernel ($\phi = 0$ on H, 1 off H), so index-2 subgroups $\leftrightarrow$ surjections $F \to \mathbb{Z}/2$.

Step 3. By Lemma 11.18, homomorphisms $F \to \mathbb{Z}/2 \leftrightarrow$ maps $\{x_1, \ldots, x_n\} \to \mathbb{Z}/2$: 2^n of them, all onto except 0. So there are $2^n - 1$.

Step 4. By Theorem 14.9, each has rank $2(n-1)+1 = 2n-1$. ■